

M.Tech. - Time Series Forecasting

Question 1: Fundamentals and Applications (20 Marks)

- (a) Define a time series. Discuss two non-obvious applications of time series forecasting outside business and stock markets (e.g., environmental, health monitoring). Explain why forecasting is critical in these contexts. (6 Marks)
- (b) A hospital records the daily number of patients visiting the emergency department over 12 days: 45, 50, 48, 55, 52, 58, 60, 62, 65, 63, 67, 70
 - (i) Compute a 3-day central moving average.
 - (ii) Using the central moving average, forecast the number of patients for day 13.(7 Marks)
- (c) Compare and contrast univariate and multivariate time series models in terms of data requirements, complexity, and interpretability. Provide an example where using multivariate modeling could be misleading. (7 Marks)

Question 2: Components and Decomposition (20 Marks)

- (a) Explain trend, seasonality, cyclic, and irregular components of a time series. Provide novel real-world examples for each, distinct from stock or sales data. (6 Marks)
- (b) You are analyzing weekly traffic flow data of a smart city sensor network. Traffic shows a rising trend and weekly seasonality. Describe step-by-step how to decompose this series using additive and multiplicative methods. Suggest which model is likely more appropriate and justify. (7 Marks)
- (c) Critically discuss a limitation of classical decomposition in detecting changing seasonal patterns over time, and propose a modern approach to overcome it. (7 Marks)

Question 3: Exponential Smoothing (20 Marks)

- (a) Discuss Simple Exponential Smoothing (SES) and explain why it is not suitable for datasets with trend or seasonality. Provide a practical example outside business sales. (5 Marks)
- (b) The weekly energy consumption (in MWh) of a small industrial plant for 6 weeks is: 500, 520, 530, 550, 560, 580
Using Holt's Double Exponential Smoothing with:

- Initial level $l_0 = 500$
- Initial trend $b_0 = 20$
- $\alpha = 0.3, \beta = 0.2$

Compute the forecast for week 7. Show all intermediate steps. (10 Marks)

- (c) A renewable energy company wants to forecast solar power generation, which exhibits both trend and complex seasonal patterns (daily and yearly). Explain why Holt-Winters triple exponential smoothing is preferred over Holt's method in this case. (5 Marks)

Question 4: Python Implementation and Critical Thinking (20 Marks)

- (a) Write a Python function to compute a weighted moving average for a given time series and a weight vector. Include comments explaining the code. (5 Marks)
- (b) A time series contains the following monthly temperature anomalies (in °C): 0.2, 0.4, 0.3, 0.5, 0.7, 0.6, 0.8, 0.9

Using Simple Exponential Smoothing with $\alpha = 0.5$, compute the forecast for the next month. Explain how changing α would impact responsiveness to sudden changes in temperature. (7 Marks)

- (c) A company must choose between 3-month moving average and Holt-Winters triple exponential smoothing for forecasting daily website traffic, which has irregular spikes during promotions. Critically discuss which method is better suited and why, considering responsiveness and seasonal patterns. (8 Marks)

Question 5 (20 Marks)

- (a) What is a time series? Differentiate between univariate and multivariate time series with suitable examples. Explain why forecasting is essential for business and economic planning. (6 Marks)
- (b) Describe the four primary components of a time series: Trend, Seasonality, Cyclical, and Irregular variations. Provide a real-world example for each component to illustrate your explanation. (6 Marks)
- (c) The following table shows the number of units (in thousands) of a product sold by a company for the first 10 months of a year:

Month	Units Sold ('000s)
Jan	24
Feb	26
Mar	23
Apr	28
May	30
Jun	25
Jul	27
Aug	32
Sep	29
Oct	28

Calculate the 3-month moving average for the given data and then forecast the sales for November. (8 Marks)

Question 6 (20 Marks)

- (a) Distinguish between the Additive and Multiplicative models for time series decomposition. Explain the underlying assumptions of each model and describe the conditions under which you would choose one over the other. (6 Marks)
- (b) Consider the following weekly sales data for a product:

Week	Sales
1	150
2	155
3	162
4	158
5	165

Using Simple Exponential Smoothing (SES) with a smoothing constant α of 0.4, find the forecast for week 6. Assume the forecast for week 1 was 150. Show your calculations for each step. (8 Marks)

- (c) Explain the primary limitation of Simple Exponential Smoothing. How does Holt's Double Exponential Smoothing model address this limitation? Describe the two smoothing equations used in Holt's model. (6 Marks)

Question 7 (20 Marks)

- (a) What is the Holt-Winters model (Triple Exponential Smoothing)? Describe its three components and explain in what type of time series data this model is most effective for forecasting. (5 Marks)
- (b) Analyse the time series plot below which shows the quarterly revenue for a retail company over 5 years. Identify and describe the key features present in the data (e.g., trend, seasonality, etc.). Justify your observations based on the plot. (5 Marks)

- (c) You are given a time series with an initial level (l_0) of 100 and an initial trend (b_0) of 5. Using Holt's Double Exponential Smoothing with $\alpha = 0.3$ and $\beta = 0.2$, forecast the value for period 4 (F_4). The observed values for the first three periods are:

- $Y_1 = 107$
- $Y_2 = 112$
- $Y_3 = 120$

Show the step-by-step calculation of the level (l_t) and trend (b_t) for each period. (10 Marks)

Question 8 (20 Marks)

- (a) Define a time series. Explain why forecasting is necessary in business and scientific applications. Provide at least two real-world examples. (5 Marks)
- (b) Discuss the differences between univariate and multivariate time series with suitable examples. Highlight one advantage and one limitation of each. (7 Marks)
- (c) A monthly sales dataset shows the following values for the first six months: [120, 135, 150, 160, 145, 155]
- (i) Compute the 3-month moving average.
 - (ii) Comment briefly on the smoothed series compared to the original.
- (8 Marks)

Question 9 (20 Marks)

- (a) Explain the different components of a time series: trend, seasonality, and irregular component. Illustrate with examples from real-world data (e.g., retail, weather, or stock market). (6 Marks)
- (b) Differentiate between the additive and multiplicative decomposition models. Provide one practical situation where each model would be appropriate. (7 Marks)
- (c) Describe a business scenario where Holt-Winters triple exponential smoothing would be a more appropriate forecasting method than Holt's double exponential smoothing. Explain what additional component the Holt-Winters method accounts for. (7 Marks)

Question 10 (20 Marks)

- (a) A company records annual demand for a product as follows: [$Y_1 = 200$, $Y_2 = 220$, $Y_3 = 240$] Using Simple Exponential Smoothing with smoothing parameter $\alpha = 0.3$, and initial forecast $F_1 = 200$, compute forecasts for Year 2, Year 3, and Year 4. Show calculations step by step. (10 Marks)

(b) You are given the following initial values for Holt's Double Exponential Smoothing:

- Initial level $l_0 = 100$
- Initial trend $b_0 = 5$
- Parameters: $\alpha = 0.3, \beta = 0.2$
- Observed values: $Y_1 = 107, Y_2 = 112, Y_3 = 120$

- (i) Update the level (l_t) and trend (b_t) for the first three periods.
- (ii) Forecast the value for Period 4.

(10 Marks)